The Journal of Open Source Software

crystALL: Purely data-driven automated structure prediction for unidentified chemical compounds

Adam M. Krajewski [1¶], **Ricardo Amaral** [1], **Jonathan W. Siegel** [2], **Rushi Gong** [1], **and Zi-Kui Liu** [1]

1 Department of Materials Science and Engineering, The Pennsylvania State University, USA 2 Department of Mathematics, Texas A&M University, USA ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Open Journals

Reviewers:

- @openjournals

Submitted: 01 January 1970
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

- (2 sentences) Motivation for materials discovery
- Typically done through combinatoric search based on protoypes and DFT
  - Proposing new structures for compounds is not a new idea based on geometrical constraints (Bail, 2005)
  - AFLOW Database (Toher et al., 2018) built by expert-intuition driven population of AFLOW Prototype Library (Mehl et al., 2016) with new chemistries in a systematic way
  - Similarity driven prediction of substitutions (Wang et al., 2021)
  - Compositional complexity grows quickly with dimensionality, considering thhat substitutions (A. M. Krajewski et al., 2024)
- (3 sentences) Past deployment on the Nd-Bi system (Im et al., 2022) and Figure 7a of (Im et al., 2022) showing excellent ordering of the candidates when evaluated vs different DFT methods. Past deployment in Al-Fe (Shang et al., 2021).

Statement of need

(statement of need)

- Motivation slightly reiterating the summary with more focus on impact and need
- (**Rushi**, 3 sentences) Importance of Ru-Zn in the study of catalysts. Importance of KCl-LaCl3 system in the study of molten salts. How `crystALL` is continuously used in the lab to predict structures of unknown compounds and if the automated experience has been positive.
- (**Adam**, 1 sentence) Conclusion on the importance of automation for user tool.

Methods

MPDD Database

Contains: - OQMD (Shen et al., 2022) - AFLOW Database (Toher et al., 2018) - Materials Project (Jain et al., 2013) - NIST-JARVIS (Choudhary et al., 2020) - COD (Gražulis et al., 2019) - GNoME (Merchant et al., 2023)

API: - native pymongo - OPTIMADE (Evans et al., 2024)

- use of `pymatgen` (Ong et al., 2013) for data processing

DRAFT

Predictive Modeling Pipeline

(about the pySIPFENN, MPDD-ALIGNN, and other models)

- pySIPFENN software (A. M. Krajewski et al., 2022) and models on Zenodo (A. Krajewski et al., 2020)
- ALIGNN (Choudhary & DeCost, 2021)

(schematic of crystALL software)

Compound-Based Structure Prediction

(about the structure prediction)

(compound figure)

Composition-Based Structure Prediction

(uncertainty in compound composition extends the previous method)

Acknowledgements

- (a loooot of grants over 5 years: NSF superalloys, PSU Seed, all from Nd-Bi study, DOE ARPA-E, NSF-POSE, ONR FGM)
- All Phases Research Lab members who tested the software in production
- All Phases Research Lab members who provided feedback on the software

Bail, A. L. (2005). Inorganic structure prediction with GRINSP. *Journal of Applied Crystallography*, *38*, 389–393. https://doi.org/10.1107/S0021889805002384

Choudhary, K., & DeCost, B. (2021). Atomistic line graph neural network for improved materials property predictions. *Npj Computational Materials*, *7*, 185. https://doi.org/10.1038/s41524-021-00650-1

Choudhary, K., Garrity, K. F., Reid, A. C. E., DeCost, B., Biacchi, A. J., Walker, A. R. H., Trautt, Z., Hattrick-Simpers, J., Kusne, A. G., Centrone, A., Davydov, A., Jiang, J., Pachter, R., Cheon, G., Reed, E., Agrawal, A., Qian, X., Sharma, V., Zhuang, H., … Tavazza, F. (2020). The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design. *Npj Computational Materials*, *6*, 1–13. https://doi.org/10.1038/s41524-020-00440-1

Evans, M. L., Bergsma, J., Merkys, A., Andersen, C. W., Andersson, O. B., Beltrán, D., Blokhin, E., Boland, T. M., Balderas, R. C., Choudhary, K., Díaz, A. D., García, R. D., Eckert, H., Eimre, K., Montero, M. E. F., Krajewski, A. M., Mortensen, J. J., Duarte, J. M. N., Pietryga, J., … Armiento, R. (2024). *Developments and applications of the OPTIMADE API for materials discovery, design, and data exchange*. https://doi.org/10.48550/arXiv.2402.00572

Gražulis, S., Merkys, A., Vaitkus, A., Chateigner, D., Lutterotti, L., Moeck, P., Quiros, M., Downs, R. T., Kaminsky, W., & Bail, A. L. (2019). Crystallography open database: History, development, and perspectives. In *Materials Informatics* (pp. 1–39). Wiley. https://doi.org/10.1002/9783527802265.ch1

Im, S., Shang, S. L., Smith, N. D., Krajewski, A. M., Lichtenstein, T., Sun, H., Bocklund, B. J., Liu, Z. K., & Kim, H. (2022). Thermodynamic properties of the nd-bi system via emf measurements, DFT calculations, machine learning, and CALPHAD modeling. *Acta Materialia*, *223*, 117448. https://doi.org/10.1016/J.ACTAMAT.2021.117448

Jain, A., Ong, S. P., Hautier, G., Chen, W., Richards, W. D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., & Persson, K. A. (2013). Commentary: The materials project:

DRAFT

A materials genome approach to accelerating materials innovation. In *APL Materials* (Vol. 1, p. 011002). American Institute of Physics Inc. https://doi.org/10.1063/1.4812323

Krajewski, A. M., Beese, A. M., Reinhart, W. F., & Liu, Z.-K. (2024). *Efficient generation of grids and traversal graphs in compositional spaces towards exploration and path planning exemplified in materials*. https://doi.org/10.48550/arXiv.2402.03528

Krajewski, A. M., Siegel, J. W., Xu, J., & Liu, Z. K. (2022). Extensible structure-informed prediction of formation energy with improved accuracy and usability employing neural networks. *Computational Materials Science*, *208*. https://doi.org/10.1016/J.COMMATSCI.2022.111254

Krajewski, A., Siegel, J., Xu, J., & Liu, Z.-K. (2020). *Neural networks for structure-informed prediction of formation energy (employed in SIPFENN)*. Zenodo. https://doi.org/10.5281/zenodo.4006802

Mehl, M. J., Hicks, D., Toher, C., Levy, O., Hanson, R. M., Hart, G., & Curtarolo, S. (2016). *The AFLOW library of crystallographic prototypes*. http://arxiv.org/abs/1607.02532

Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *80 | Nature |*, *624*. https://doi.org/10.1038/s41586-023-06735-9

Ong, S. P., Richards, W. D., Jain, A., Hautier, G., Kocher, M., Cholia, S., Gunter, D., Chevrier, V. L., Persson, K. A., & Ceder, G. (2013). Python materials genomics (pymatgen): A robust, open-source python library for materials analysis. *Computational Materials Science*, *68*, 314–319. https://doi.org/10.1016/j.commatsci.2012.10.028

Shang, S.-L., Sun, H., Pan, B., Wang, Y., Krajewski, A. M., Banu, M., Li, J., & Liu, Z.-K. (2021). Forming mechanism of equilibrium and non-equilibrium metallurgical phases in dissimilar aluminum/steel (al–fe) joints. *Scientific Reports*, *11*, 24251. https://doi.org/10.1038/s41598-021-03578-0

Shen, J., Griesemer, S. D., Gopakumar, A., Baldassarri, B., Saal, J. E., Aykol, M., Hegde, V. I., & Wolverton, C. (2022). Reflections on one million compounds in the open quantum materials database (OQMD). *Journal of Physics: Materials*, *5*, 031001. https://doi.org/10.1088/2515-7639/ac7ba9

Toher, C., Oses, C., Hicks, D., Gossett, E., Rose, F., Nath, P., Usanmaz, D., Ford, D. C., Perim, E., Calderon, C. E., Plata, J. J., Lederer, Y., Jahnátek, M., Setyawan, W., Wang, S., Xue, J., Rasch, K., Chepulskii, R. V., Taylor, R. H., … Curtarolo, S. (2018). The AFLOW fleet for materials discovery. In *Handbook of Materials Modeling* (pp. 1–28). Springer International Publishing. https://doi.org/10.1007/978-3-319-42913-7_63-1

Wang, H.-C., Botti, S., & Marques, M. A. L. (2021). Predicting stable crystalline compounds using chemical similarity. *Npj Computational Materials*, *7*, 12. https://doi.org/10.1038/s41524-020-00481-6

DRAFT